Can the issuer's estimated value be reproduced from public data?

Evidence from worst-of autocallable notes

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Abstract

US structured note issuers print an estimated value on the cover of every pricing supplement, computed with models and a funding rate the filing does not disclose. This paper asks whether an outside party can reproduce that figure, and tests the question on the hardest instrument it covers: worst-of autocallable notes on three correlated equity indices, whose redemption before maturity is the issuer's to initiate rather than automatic on five of the six notes studied. Surfaces are fitted to public option quotes under no-arbitrage constraints, converted to local volatility, and simulated jointly, with the issuer's call solved by least-squares Monte Carlo from the issuer's side of the contract.

The disclosed figure is not reproduced. Independent valuations land below it on all six notes, by \$11.41 to \$27.92 per \$1,000 of principal across the five whose barriers are covered by quoted strikes, against simulation standard errors below \$0.40. Solving for the funding spread that would reconcile the two valuations returns a negative spread on every note, from −278 to −123 basis points against an A-rated corporate benchmark of 69, so the undisclosed funding rate cannot account for the difference. Four further candidates are excluded on direction or magnitude. Two survive, the correlation assumption and the shape of the volatility surface, and a third, the dynamics of that surface, lies outside what a local-volatility model can test; none of the three is published in the filings. A census of the same week bounds the exercise: of 1,086 supplements filed, 129 are final notes of this type, and 71 of those can be priced from public quotes at all.

Keywords: structured notes; autocallable; estimated value; local volatility; Monte Carlo.

JEL classification: G12, G13, G14, G24.

1 Introduction

The cover page of a US structured note's pricing supplement carries an estimated value: the issuer's own figure for what the security is worth, set below the price the investor pays for it. One of the notes examined here was sold at \$1,000 and disclosed at \$938.90. The practice traces to a sample comment letter from the SEC's Division of Corporation Finance (2012), which asked issuers to "consider prominently disclosing" such a figure; it is staff guidance and not a rule, and no rule has since replaced it. What the filing omits is how the number was reached. The model belongs to the issuer, so does the volatility surface, and the discount curve embeds an internal funding rate that appears nowhere in the document. An investor is given a valuation and no means of checking it.

This paper asks whether the figure can be reproduced from outside the issuer, using only public data.

The question is put to worst-of autocallable notes, which are the most demanding instrument in the disclosure rather than the most convenient. Payment tracks the weakest of three correlated equity underlyings.

A contingent coupon is paid only when every leg closes at or above its barrier on an observation date, and principal is returned in full only if the weakest leg finishes above a lower trigger. The call deserves particular attention, because the product's name misdescribes it. On five of the six notes priced here redemption is not automatic: three are redeemable at the issuer's option, and two are redeemed, in the issuer's own words, "if and only if the output of a risk neutral valuation model … indicates that redeeming on such date is economically rational for us." Only one note in the sample redeems mechanically when a level is reached. Valuing any of them calls for a joint model that carries each leg's skew, a correlation structure, and an exercise rule solved from the issuer's side of the contract, where the holder of the option is short the investor's payoff and stops to minimise it.

Every input is public. Implied volatility surfaces are fitted to delayed end-of-day option quotes from Cboe under the no-arbitrage conditions of raw SVI, converted to Dupire local volatility, and simulated jointly against a Treasury discount curve; the issuer's call is solved by Longstaff–Schwartz regression on 200,000 paths. Before any note is priced the engine is made to return the vanilla prices it was calibrated to, measured against the quoted bid-ask width rather than in isolation, since an error only counts as large relative to the price uncertainty of the market it came from. Four of the five surfaces return a median error below 0.26 of the quoted bid-ask width, well inside the band the market itself prices in. The S&P 500 surface does not, and §5 attributes that to its exceptionally narrow quoted width rather than to a poor fit: at 0.131 volatility points its absolute error falls inside the range the other four surfaces span, while its quoted width is 4.7 times narrower than the next tightest in the sample.

The answer is no. On all six notes the independent valuation lands below the disclosed figure, by $11.41 to $27.92 per $1,000 of principal across the five notes whose barriers are covered by quoted strikes, or 1.17 to 2.97 per cent of the disclosed value. Inverting the question makes the size of the problem clearer: solving for the constant funding spread over Treasuries that would set model value equal to disclosed value returns a negative spread on every note for which the solution brackets, from −278 to −123 basis points, against an option-adjusted spread of 69 basis points on A-rated US corporate debt on the pricing date. Reconciliation would require the issuer to borrow below the Treasury curve. A funding rate the filing does not disclose is therefore not a sufficient explanation for the difference, which is the candidate a reader would reach for first.

What follows from this is narrower than it may appear, and the limits are worth stating before the evidence. The result is not a finding of overpricing. Henderson and Pearson (2011) established that these notes are issued above model value, and the disclosed estimated value is itself below the issue price, so both orderings already hold; the gap measured here is between the disclosed value and an independent valuation, not between either and the price. Nor does this paper identify what the residual is made of, though it narrows the field. §7 takes the candidates one at a time and excludes four of them: funding and observation timing move the value the wrong way, while the proxied leg and the term extrapolation move the right way by too little or on too few notes. Suboptimal exercise accounts for part of three notes and is arithmetically unavailable for two. Two measured candidates survive, the correlation assumption and the shape of the volatility surface, and a third, the dynamics of that surface, is one a local-volatility model cannot test at all. None of the three is published in the filings. Which is at work, or in what combination, is not something six notes can settle. Six notes from one week support a bounded claim and no population statement.

Two contributions remain. The first is the verification exercise itself, run on the product class the nearest published work set aside: Bauer, Fink, and Stoller (2020) check issuer-disclosed values against standard option pricing models on German retail certificates and state plainly that "[m]ore complex products like autocallables are left for future research." This paper prices them, and reports the funding-spread inversion as a diagnostic that converts an unexplained difference into a statement about what the difference cannot be.

The second is a census of what any such exercise can reach. Of 1,086 prospectus supplements filed on Form 424B2 in a single week, 129 are final worst-of notes with a contingent coupon and a disclosed estimated value, and 71 of those, 55 per cent, can be priced from public option quotes at all. The remainder reference underlyings with no listed options or no usable chain. Any outside verification of this disclosure is bounded by that fraction, and §3 reports where the rest of the market goes.

2 The disclosed estimated value, and prior work

2.1 What the issuers say the number is

The filings are explicit that the estimated value is a model output and whose model it is. One of the supplements priced here states that the issue price "exceeds the estimated value of your notes as of the time the terms of your notes are set on the trade date, as determined by reference to GS&Co.'s pricing models and taking into account our credit spreads," and that those "pricing models are proprietary and rely in part on certain assumptions about future events, which may prove to be incorrect."[1] The same document lists the inputs: "our credit spreads, interest rates (forecasted, current and historical rates), volatility, price-sensitivity analysis and the time to maturity of the notes."

That list is worth reading closely, because correlation does not appear in it. The note it describes pays on the weakest of three equity indices, and §6 shows the investor is long correlation on every note in the sample. Morgan Stanley's supplements list their inputs too, naming "market inputs and assumptions relating to the underliers, instruments based on the underliers, volatility and other factors including current and expected interest rates," and correlation is absent there as well. Both lists are introduced as partial, with "including principally" and "other factors," so neither purports to be complete and no inference follows that either model ignores correlation. What does follow is narrower and is the point of this paper: the disclosure names the inputs it considers principal, and the input that most distinguishes a worst-of structure from a single-underlying one is not among them.

The gap between the issue price and the estimated value is itself attributed, in the same document, to "the underwriting discount and commissions, the expenses incurred in creating, documenting and marketing the notes, and an estimate of the difference between the amounts we pay to GS&Co. and the amounts GS&Co. pays to us in connection with your notes." The first two components appear on the cover as numbers. The third does not. It is the internal transfer between the issuing entity and its distribution affiliate, and it is the component that the phrase "internal funding rate" is usually reaching for. That phrase belongs to the issuers rather than to the guidance. The 2012 comment letter discussed below does not contain it, while JPMorgan's supplements in this sample carry a risk factor headed "Estimated Value of the Notes Is Derived by Reference to an Internal Funding Rate." The attribution matters because the rate it names is the one input a reader cannot recover.

One further sentence in the same risk factor deserves attention, since it anticipates this paper's exercise and declines to bound it. The actual value an investor could realise "may differ, perhaps materially, from the estimated value of your notes determined by reference to our models due to, among other things, any

[1] Quotations in this subsection are from the pricing supplement for note N4, filed by GS Finance Corp. under accession number 0001193125-26-419171 and preserved in full in the project repository; the filings for all six notes are identified in Appendix A. The other three issuers in the sample describe the figure in materially similar terms, each naming its own proprietary models: Morgan Stanley's supplements say the estimated value "is determined using our own pricing and valuation models" and that these "are proprietary," and BofA's say it is "determined as of the pricing date by reference to our and our affiliates' pricing models."

differences in pricing models or assumptions used by others." The filing therefore tells the reader that another party's model may disagree, by an amount it does not characterise, for reasons it does not enumerate. Putting a number on that disagreement for a specific class of notes, from inputs the reader can obtain, is what follows in §6.

2.2 Where the disclosure comes from

The practice originates in an April 2012 sample comment letter from the Division of Corporation Finance, addressed to financial institutions issuing structured notes (US Securities and Exchange Commission, Division of Corporation Finance, 2012). On the valuation figure the letter asks that issuers "should consider prominently disclosing" it, language that is advisory in form as well as in effect. A sample comment letter is staff guidance: it conveys the views of the staff, not a rule adopted by the Commission, and it binds no one. No rule has since replaced it for this figure.

The practical consequence is that there is no prescribed form, and the sample examined here shows the range that permits. Among the final worst-of contingent-coupon supplements filed in the sample week, five carry no point estimate in the filed document. Three state the figure as a bound rather than a number, with the cover reading "Our estimated value of the notes on trade date: Not less than the face amount of such notes." The remaining two carry only the risk-factor heading and locate the figure in an accompanying index supplement that is not part of the document filed under the cover.[2] All five come from one issuer. None of this is a deficiency, since nothing requires a point estimate; it is what discretionary disclosure looks like when it is exercised.

The timing is constrained even where the content is not. Rule 424(b)(2) requires the priced supplement to be filed no later than the second business day following the earlier of the date of determination of the offering price or the date it is first used (US Securities and Exchange Commission, 2024). A verification exercise therefore has a short and known window in which the terms are fixed and public, which is what makes a single week a coherent sample rather than an arbitrary one.

2.3 Prior work

The literature on retail structured products has measured issuer margins for two decades, mostly in Europe and mostly by replication. Wilkens, Erner, and Röder (2003) document premia over replication cost and a life-cycle pattern in German products; Stoimenov and Wilkens (2005) extend the fair-pricing analysis across equity-linked retail instruments; Entrop, Fischer, McKenzie, Wilkens, and Winkler (2016) quantify overpricing of discount certificates and attribute a substantial share of it to issuer credit risk; and Ruf (2011) relates pricing power to credit quality. In the US, Henderson and Pearson (2011) is the canonical study, establishing that these notes are issued above model value, and Deng, Mallett, and McCann (2011) give an early valuation treatment of autocallable structures alongside evidence of their growth. The methods for valuing the structures themselves are well developed: Guillaume (2015) derives analytical results covering memory coupons, partial protection and worst-of contingencies, and Kim and Lim (2019) construct static replications of autocallable and barrier features as an alternative to simulation.

[2]These five supplements are not among the six priced here and are therefore not preserved in the project repository, which holds only the documents the valuations depend on. They remain retrievable from EDGAR under accession numbers 0001193125-26-415466, 0001193125-26-413990 and 0001193125-26-413989 for the three quoting a bound, and 0000950103-26-015173 and 0000950103-26-015172 for the two deferring to a supplement.

This paper differs from all of these in its object. They measure the distance between a price and a model value. The question here is the distance between two model values, one of which the issuer published. The nearest precedent for treating a disclosed figure as the thing under examination is Jørgensen, Nørholm, and Skovmand (2011), who compare overpricing against costs the issuer disclosed at issuance and find roughly half of it unexplained. On the US estimated-value disclosure specifically, Lai (2022) assembles a large EDGAR sample of auto-callable reverse convertibles, but studies realised returns rather than whether the disclosed figure can be reproduced.

The closest published work is Bauer et al. (2020), who check issuer-disclosed fair values against standard option pricing models for German retail certificates and conclude that the disclosed values can be reproduced. Their sample is of simpler instruments, and they say so: "[m]ore complex products like autocallables are left for future research." Three features of the setting examined here make that an unusually clean gap to occupy. The product is a worst-of on several correlated underlyings rather than a single one. On five of the six notes redemption before maturity is not automatic, three being at the issuer's option and two following the disclosed rational-exercise rule quoted in §1, so valuation requires an optimal-stopping problem solved against the investor rather than a barrier check. And the figure rests on staff guidance that prescribes neither a method nor a form, which is what makes reproducing it a question about the disclosure and not only about the model. Whether a disclosed value can be reproduced is therefore open in this market rather than settled elsewhere and carried across.

3 Sample and data

3.1 Why the sample is one week

A note can only be repriced against its disclosed value while its terms are both fixed and public, and that state begins later than one might expect and is announced only briefly. The preliminary supplement carries ranges: a coupon quoted as "at least" some rate, a trigger to be determined, an estimated value given as an interval.[3] The priced supplement replaces all of it with numbers, and Rule 424(b)(2) requires that document to be filed no later than the second business day following the earlier of the date the offering price was determined or the date the supplement is first used (US Securities and Exchange Commission, 2024).

So the terms arrive in a stream of final supplements with a two-day lag and no subsequent amendment. Choosing a calendar week and taking every qualifying filing in it is therefore a sampling rule a reader can check, and one that cannot be tuned after the fact by extending the window until the results improve. The week used here is 5 to 9 October 2026. Five of the six notes were priced on 7 October and the sixth on 5 October.

3.2 What the week contained

Table 1 is the whole filing population, each filing assigned to the first stage it fails, so the rows partition rather than overlap. Of 1,086 supplements filed on Form 424B2 that week, 667 are preliminary and 285 are final but not the structure under study, either because they are not worst-of or because they carry no contingent coupon. A further five are final worst-of notes with a contingent coupon that state no point estimate of the value, discussed in §2.

[3]The phrasing is standard across the preliminary supplements in this week. One example, filed under accession number 0001918704-26-031175, offers "a Contingent Interest Rate of at least 9.70% per annum." These documents are superseded by their priced counterparts and are not preserved in the repository.

Table 1. Every 424B2 filed in the pricing week, by the reason it does or does not reach the sample

Disposition	Filings	Share (%)
Preliminary supplement; terms not yet fixed	667	61.4
Not a worst-of structure	201	18.5
No contingent coupon	84	7.7
No estimated value disclosed	5	0.5
Single-name or fund underlyings	16	1.5
An underlying with no listed options	42	3.9
Priceable from public option quotes	71	6.5
Total 424B2 filings	*1,086*	*100.0*

Notes: All 424B2 filings retrieved from EDGAR full-text search for the week of the snapshot. Each filing is assigned to the first stage it fails, so the rows partition the population. A preliminary supplement is excluded because its coupon, trigger and estimated value are still ranges. The last three rows are the final worst-of notes with a contingent coupon and a disclosed estimated value, 129 in all, of which 71 (55%) can be priced from public option quotes; the six notes studied here are drawn from those 71. Source: `src/data/edgar_notes.py`.

Table 2. The six notes: underlyings, terms and the figures the issuer discloses

Note	Issuer	Underlyings	Tenor	Coupon	Barrier	Trigger	Est. value	Commission
N1	JPMorgan	NDX, RUT, SPX	5.00	8.30	70.00	65.00	938.90	41.25
N2[a]	Morgan Stanley	DJX, NDX, RUT	5.00	8.30	70.00	65.00	943.10	36.25
N3	Morgan Stanley	NDX, RUT, SPX	3.01	9.75	70.00	60.00	983.40	2.50
N4	Goldman	NDX, RUT, SPX	1.92	10.15	65.00	60.00	999.00	5.25
N5	BofA	NDX, RUT, SPX	1.75	6.25	75.00	70.00	972.10	25.00
N6	JPMorgan	RUT, SPX, SMH	1.92	12.40	65.00	60.00	967.30	22.25

Notes: Tenor in years from pricing date to stated maturity. Coupon is the contingent rate per annum in per cent, paid only when every underlying is at or above its coupon barrier on an observation date. *Barrier* is that coupon threshold and *Trigger* the principal-repayment threshold, both as a percentage of the initial level. Estimated value and selling commission are per $1,000 of principal, as disclosed in the pricing supplement. [a] Held out of the ranges quoted in the text: no fitted slice past one year carries a quoted strike at this note's trigger on one leg, so its downside is extrapolated rather than observed. Source: the filings listed in Appendix A.

That leaves 129 final worst-of contingent-coupon notes with a disclosed estimated value, of which 71 can be priced from public option quotes. Of the 58 that cannot, 16 are written on single names or funds and 42 carry at least one leg whose options are not listed. That second group is more concentrated than one would guess. In 33 of the 42 the only such leg is the Nasdaq-100 Technology Sector index, a US index with no options market of its own; the remaining nine reference the EURO STOXX 50, the Nikkei 225 or a futures-based Russell variant. The ratio is worth carrying forward. An outside party attempting this verification can reach 55 per cent of the qualifying market, and what it cannot reach is not a random sample of the rest but is dominated by notes on a single index that has no options market of its own.

3.3 The six notes

Six of the 71 are priced here, listed in Table 2, chosen to span the structural variation rather than to be representative in any statistical sense. They come from four issuers. Tenors run from 1.75 to 5.00 years, contingent coupons from 6.25 to 12.40 per cent per annum, coupon barriers from 65 to 75 per cent of the initial level and downside triggers from 60 to 70 per cent. The disclosed estimated values run from $938.90

to $999.00 per $1,000 and the selling commissions from $2.50 to $41.25, the low figure belonging to a note sold only into fee-based advisory accounts where the adviser is compensated separately.

The call mechanics vary more than the name of the product suggests, and the variation is the reason six notes are enough to be interesting. Three are redeemable at the issuer's option on scheduled dates. Two redeem under the disclosed rational-exercise rule quoted in §1. One, and only one, is auto-callable in the ordinary sense: redemption is mechanical when every underlying closes at or above 93.25 per cent of its starting level on one of six observation dates. That threshold is itself worth noting, since discussions of these products often assume the automatic call sits at the initial level.

3.4 Option data and the fitted surfaces

Implied volatility surfaces are fitted to delayed end-of-day option chains retrieved from Cboe's public endpoint and preserved as of 9 October 2026, with a SHA-256 digest recorded for every file. Five surfaces cover the eighteen leg positions the six notes carry between them, which resolve to five distinct underlyings. Expiries inside a fortnight are dropped, as are quotes whose relative bid-ask exceeds 0.35, and log-moneyness is clipped to $[-0.9, 0.45]$; consecutive expiries closer than 25 days apart are thinned so that the calendar condition is tested across separated maturities rather than across adjacent weeklies.

Each slice's forward and discount factor come from a put-call parity regression across near-the-money strikes. Three of the five surfaces are fitted to cash index options, which are European, and for these both quantities are fitted. The other two are fitted to exchange-traded fund options, which are American, and there the discount factor is pinned to the Treasury curve and only the forward is fitted. Parity is an identity only under European exercise: left to fit freely on the semiconductor ETF, the regression returned a discount factor implying a 1.6 per cent zero rate against 4.3 per cent Treasuries, because the early-exercise premium on in-the-money puts grows with strike and tilts the slope. Pinning the discount factor removes that tilt while leaving the forward to carry the market's dividend assumption. Near the money the early-exercise premium itself is small enough to leave in the quotes, which is an approximation rather than a correction and is revisited in §8.

3.5 The one proxied leg

Four of the five underlyings are priced off options on the instrument named in the term sheet. The exception is the Dow Jones Industrial Average, whose listed index options are too sparse to fit a surface across the maturities these notes need, and which is therefore priced off options on the ETF that tracks it.

What that substitution costs can be measured rather than assumed, because two other legs in the sample have both an index and an ETF chain available at shared expiries. At-the-money implied volatility is higher on the fund than on the index in every comparison: by 0.517 volatility points for the Dow pair at the one expiry they share, and by 0.434 and 0.435 points on average for the Nasdaq-100 and Russell 2000 pairs across 16 shared expiries each, reaching beyond one and two years respectively. The sign is consistent and the magnitude is similar across all three, which is what makes the Dow figure usable as an estimate rather than a single observation.

Correcting that one leg, and nothing else, shifts the affected note's value from $893.93 to $896.54, a rise of $2.61 with a standard error of $0.73, and narrows its difference from the disclosed value from $49.17 to $46.56. The proxy therefore accounts for about five per cent of that note's gap and moves it in the direction that would close it. It is the same note whose downside trigger falls outside the quoted strike range on the

Dow leg, which is why §6 reports it but holds it out of the stated ranges; Appendix C gives the coverage counts leg by leg.

3.6 Rates, spreads and the volatility indices

Discounting uses the US Treasury par yield curve for the valuation date, as published by the Treasury. The benchmark against which an implied funding spread is judged is the ICE BofA A-rated US corporate index option-adjusted spread, 69 basis points on the pricing date, taken from the Federal Reserve Bank of St Louis. The Cboe volatility indices for the S&P 500, Nasdaq-100 and Russell 2000 supply the surface move between the pricing date and the snapshot date used in §7, each index matched to the leg it measures. Daily closes for the semiconductor ETF come from a commercial endpoint rather than an official series, no free official source for a single fund's history being available; that series affects one note of the six.

Every series above is free and unauthenticated, and the retrieval code is in the repository. The option chains the reported results depend on are preserved with their digests, since a delayed-quote endpoint serves only current quotes and the 9 October snapshot cannot be retrieved again.

4 Method

4.1 The volatility surface

Each expiry is fitted separately in total implied variance against log-moneyness, $w(k) = \sigma_{\mathrm{BS}}^2(k)\,T$ with $k = \ln(K/F_T)$, using the raw SVI parameterisation of Gatheral and Jacquier (2014),

$$w(k) = a + b\left[\rho\,(k-m) + \sqrt{(k-m)^2 + s^2}\,\right]. \tag{1}$$

Fitting variance rather than volatility is a choice the next step forces. Local volatility needs $\partial w/\partial T$ and $\partial^2 w/\partial k^2$, and (1) is twice differentiable in k in closed form, so those derivatives come off a fitted slice instead of off differenced quotes.

Two arbitrage conditions are imposed, both from Gatheral and Jacquier (2014). The risk-neutral density implied by a slice is non-negative exactly when

$$g(k) = \left(1 - \frac{k\,w'}{2w}\right)^2 - \frac{(w')^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2} \;\geq\; 0 \tag{2}$$

everywhere, and total variance must not decrease in T at fixed k, or a calendar spread prices negative. Neither check is cosmetic. A slice violating (2) has negative density somewhere, and the denominator of the local-variance expression below passes through zero at that point, which is the usual reason a local-volatility surface comes back full of undefined values. When a check fails the slice is refitted subject to the constraint rather than repaired afterwards, the arbitrage-free parameter domain being characterised by Martini and Mingone (2022).

Where the conditions are enforced turns out to matter as much as whether they are. Checking on one fixed wide grid asks a two-week slice about strikes far below its deepest quote, where it is purely extrapolating, and enforcing the calendar condition out there makes each short-dated slice's extrapolated wing a floor for the next expiry, so the constraint propagates forward from the slice carrying the least information rather than the most. The grid is scoped to the strikes each pair of adjacent expiries actually quotes, padded slightly and capped at

five standard deviations of total volatility. Quotes enter the fit only if the expiry is at least a fortnight out and the relative bid-ask is at most 0.35, with log-moneyness clipped to $[-0.9, 0.45]$, and consecutive expiries closer than 25 days apart are thinned so that the calendar condition is tested across separated maturities rather than adjacent weeklies. The surfaces retain between 11 and 20 fitted slices each.

4.2 Local volatility

The simulation diffusion is Dupire local volatility, taken in Gatheral's log-moneyness form,

$$\sigma_{\mathrm{LV}}^2(k,T) = \frac{\partial w/\partial T}{1 - \dfrac{k}{w}\dfrac{\partial w}{\partial k} + \dfrac{1}{4}\left(-\dfrac{1}{4} - \dfrac{1}{w} + \dfrac{k^2}{w^2}\right)\left(\dfrac{\partial w}{\partial k}\right)^2 + \dfrac{1}{2}\dfrac{\partial^2 w}{\partial k^2}}. \tag{3}$$

Derivatives in k are exact. In T the surface is interpolated linearly in total variance between adjacent expiries, which makes $\partial w/\partial T$ piecewise constant but non-negative, because the calendar condition already forced w to be non-decreasing in T. The butterfly condition keeps the denominator of (3) positive. Local variance is therefore positive by construction rather than by inspection, which is the reason the two checks in the previous subsection are not optional. The implementation still counts how often it has to clamp, so a failure would be visible rather than silent.

Past the longest listed expiry the surface must be extrapolated, and for the five-year notes that is a long way. Quoted expiries reach 5.20 years on the S&P 500 but stop at 3.20 on the Nasdaq-100 and 2.19 on the Russell 2000, while the two longest notes run to 5.00 years. The base case holds implied volatility flat in maturity with its smile shape intact, $w(k,T) = w_{\mathrm{last}}(k)\,T/T_{\mathrm{last}}$. Two alternatives are implemented so the choice can be measured rather than asserted: carrying the last observed variance slope forward, and shifting every volatility by a fixed amount. §7 reports what the choice is worth, which ranges from \$0.22 on the shortest note to \$12.36 on the longest.

4.3 Correlation

Correlation is estimated from 252 trading days of daily log returns, taken up to and including each note's own pricing date rather than the snapshot date. Five of the six notes priced two business days before the snapshot, so using everything available would feed future returns into a historical pricing decision; the effect on the matrix is small and the principle is not. Where the sample matrix is not positive semidefinite, the nearest valid correlation matrix in the Frobenius norm is substituted by the alternating-projection method of Higham (2002).

This is the largest approximation in the method and it runs in a known direction. Vanilla options carry no information about co-movement, and there is no free source of implied correlation, so a realised estimate stands in for a risk-neutral input. Dealers mark correlation above realised for precisely this product, because selling a worst-of leaves the desk short correlation. The investor is long correlation on every note in the sample, so if the issuer used a higher figure than the one estimated here, its value would exceed this paper's for that reason alone. That makes correlation the first candidate §7 has to rule in or out, and the one it treats at most length.

4.4 Simulation

Paths are generated by daily log-Euler steps under the risk-neutral measure, each asset drifting along its own forward curve so that carry is whatever the option market implies rather than a separately assumed dividend yield, and diffusing at (3). Shocks are correlated through the Cholesky factor of the correlation matrix. Antithetic variates are used, which help most where the payoff is discontinuous, and the reported figures use 200,000 paths.

Three implementation choices are worth recording because they decide what is computationally possible rather than merely convenient. Paths are not stored: a five-year note observed monthly needs 1,260 daily steps on three assets, and retaining 200,000 of them would run to several terabytes, so the engine carries only the current level and records at each observation date the statistics the payoffs need, which are the worst and best performances across underlyings. The American knock-in is handled by a running indicator updated every day, since the condition it tests, that an underlying's value falls below its threshold "on any Trading Day during the Knock-In Period," cannot be evaluated from the observation dates alone. That note's loss condition is a conjunction rather than a single test: a knock-in must occur and the least performing underlying must also finish below its starting value, not merely below the threshold that triggered the knock-in. Local volatility is read from a precomputed grid in (k, T) with bilinear interpolation rather than evaluated analytically at every step, which reproduces the surface to well under a basis point of volatility and is what makes bump-and-revalue Greeks affordable. Common random numbers come from re-seeding rather than from stored draws, so a bumped revaluation differs from the base case only in the bumped input and each Greek is a clean finite difference.

A European control variate is applied, with the variance reduction factor running from 1.8 to 3.8 across the six notes. Standard errors on the reported values fall between \$0.24 and \$0.37 per \$1,000.

4.5 The issuer's call

The Bermudan call is solved by least-squares Monte Carlo (Longstaff & Schwartz, 2001). Treating an issuer's early-redemption right as an optimal-stopping problem rather than as a barrier test has published precedent in this product family: Detemple and Kitapbayev (2021) solve for the optimal policy explicitly on callable barrier reverse convertibles, and Huo (2025) applies least-squares Monte Carlo to worst-of issuer-callable notes of the kind priced here, under a multi-variate local-volatility model and with the stopping problem posed from the issuer's side. One sign convention has to be stated plainly, because reversing it would change every result. A high continuation value is the issuer's liability, not its asset. The issuer therefore redeems when keeping the note alive costs it more than repaying principal now, so the investor's value at a call date is the redemption amount whenever the estimated continuation value exceeds it. Solved the other way the issuer's option becomes a gift to the investor and every price rises.

The regression basis is deliberately small: a constant, the worst performance W, its square, the spread between best and worst performance, and accrued-but-unpaid coupon. The worst performer is the state variable that drives the payoff, and its square lets the exercise boundary curve; the best-worst spread carries the dispersion a worst-of note is exposed to; accrued coupon matters only where the note has a memory feature. A richer basis fits simulation noise and estimates a worse boundary rather than a better one.

One detail of the comparison is easy to get wrong in the direction the term sheets invite. A coupon due on a call date is paid whether or not the note is redeemed, so it cancels from the decision and the continuation value is compared against par alone. Term sheets quote the redemption as par plus coupon, and carrying

that wording into the comparison makes calling look about one coupon more expensive than it is, which systematically under-calls and overvalues the note.

4.6 The funding-spread inversion

The gap between a model value and the disclosed value can be expressed as the discount-rate adjustment that would remove it. For a constant spread s over the Treasury zero curve,

$$V(s) = \mathbb{E}^{\mathbb{Q}}\left[\sum_i e^{-(r_{t_i}+s)\,t_i}\,\mathrm{CF}_i\right], \qquad \text{solve } V(s^*) = \mathrm{EV}_{\text{disclosed}}. \tag{4}$$

The spread enters the exercise policy as well as the discounting, so (4) is solved by repricing on a fixed set of paths rather than by rescaling a single valuation. V is monotone decreasing in s, so the root is bracketed and found by Brent's method.

What makes s^* informative is that its sign is known in advance. Discounting a liability at a wider spread lowers its value, and an issuer funds above Treasuries, so a disclosed value derived with the issuer's own funding rate should reconcile at a positive s^*. A negative solution cannot be a funding spread. Because the issuer's call is the largest discretionary input in the model, (4) is solved under two assumptions about it, optimal exercise and no early call, and §6 reports both.

5 Validation

A note's value is not observable, so nothing downstream of the pricing engine can be checked against a market. Everything upstream can be, and is, because a model that cannot return the instruments it was calibrated to has no claim on the instrument it was built for. This section reports those checks in the order the pipeline runs them, including the one that does not pass.

5.1 Forwards and discount factors

Each slice's forward comes from a put-call parity regression across near-the-money strikes. The direct check is how well the fitted forward agrees with the one the quotes imply, and the maximum relative error across expiries is 1.6, 1.9 and 4.4 basis points on the three cash indices, 8.0 on the fund standing in for the Dow and 16.7 on the semiconductor ETF. The two largest are the two exchange-traded funds, for the reason §3 gives: their options are American, parity is an identity only under European exercise, and the discount factor is therefore pinned to the Treasury curve rather than fitted.

Where the discount factor is fitted, it is worth reading as a diagnostic in its own right. The implied zero rates run from 3.73 to 5.47 per cent and rise with maturity, against a Treasury curve near 4.3 per cent over the same horizon. The short end is noisy, which is expected when the regression has a few weeks of time value to work with. The long end is not noise: index options embed a financing rate that sits above Treasuries, and the fit recovers it. That rate enters the forward, and so the diffusion, but not the discounting of the note's cashflows, which is a separate curve and the object §6 solves for.

Table 3. Repricing error against quoted bid-ask width, by surface, in implied volatility points

Surface	Options	Inside spread (%)	Median error	Median spread	Error/spread
DJX	518	96.3	0.067	0.928	0.07
NDX	1,074	98.6	0.111	0.561	0.20
RUT	845	96.0	0.097	0.368	0.26
SMH	921	99.5	0.173	2.157	0.08
SPX	2,378	46.3	0.131	0.079	1.67

Notes: Every option used in the calibration is repriced through the local-volatility engine and the result converted back to implied volatility, so the error is in the units the market quotes. *Inside spread* is the share of repriced options falling within the quoted bid-ask. The final column is the median error divided by the median quoted bid-ask width. Error is measured from the mid and the quoted band runs half a width either side of it, so a ratio below one half puts the model inside the band the market itself prices in. Four surfaces return 0.26 or better; the SPX surface returns 1.67, on a quoted width 4.7 times narrower than the next tightest here, which §5 attributes to that market's liquidity rather than to the fit. Source: `src/vol/reprice_check.py`.

5.2 The arbitrage audit

Every fitted slice is checked on its own grid for both conditions of §4. Across all five surfaces and their 68 slices there are zero butterfly failures and zero calendar failures. The minimum of $g(k)$ is positive on every surface, at its tightest 0.0057 on the S&P 500, and the implied risk-neutral density is positive everywhere.

The calendar diagnostic is the more informative of the two. On three of the five surfaces the minimum gap between adjacent total-variance curves sits at the solver's tolerance, around 10^{-6}, which means the constraint is binding rather than slack: those slices were refitted against it, and without the constraint the surface would have violated it. On the other two the gap is three orders of magnitude larger and the condition holds on its own. Reporting the audit as a pass alone would hide that difference.

5.3 The repricing gate

The gate is whether the local-volatility engine, built from a surface, returns the prices of the vanilla options that surface was fitted to. Each option used in calibration is repriced through the full Monte Carlo machinery and the result converted back to implied volatility, so the error is in the units the market quotes rather than in dollars, where it would be dominated by the level of whichever index it came from. Table 3 reports it.

An error is only large or small relative to the price uncertainty of the market it came from, so the comparison is against the quoted bid-ask width rather than against zero. Error is measured from the mid and the quoted band runs half a width either side, so a ratio below one half puts the model inside the band the market itself prices. Four of the five surfaces clear that comfortably, at 0.07, 0.08, 0.20 and 0.26. Between 96.0 and 99.5 per cent of their repriced options land inside the quoted bid-ask outright.

Figure 1 shows what that looks like for six S&P 500 expiries, with the per-slice fit error in basis points. The fits track the quoted mids through the body of the smile and part company in the upper wing on two of the panels, where the quotes thin out. The last panel is the longest listed expiry at 5.20 years, fitted from ten quoted volatility points against 49 on the expiry a year before it. Nothing about the extrapolation discussed in §4 shows up as error here, because by that maturity there is very little left to be in error against.

5.4 The surface that does not pass

The S&P 500 surface returns a ratio of 1.67, and 46.3 per cent of its repriced options fall inside the quoted bid-ask against 96 per cent or better everywhere else. Taken at face value that is the worst-fitted surface in the study, and it belongs to the most liquid index option market in the world, which should be reason enough to look at the denominator before accepting the verdict.

Its absolute error is 0.131 volatility points. That is the second largest of the five, sitting between the Nasdaq-100 at 0.111 and the semiconductor ETF at 0.173, and inside the range the other four span. What distinguishes it is the width it is divided by. The median quoted bid-ask on the S&P 500 slices is 0.079 volatility points, against 0.368 on the next tightest surface and 2.157 on the widest. The S&P 500 market quotes roughly four and a half times tighter than anything else in the sample, so a fit that is unremarkable in absolute terms fails a test scaled to that market's own precision.

Two readings are available and the data does not fully separate them. Either the calibration is genuinely less accurate on this surface than the absolute error suggests, or the test is asking a surface fitted from end-of-day quotes to reproduce intraday-tight spreads that no daily snapshot could support. The second is more consistent with the evidence, since the absolute error is ordinary and the arbitrage audit on this surface is clean. It is reported either way, because a validation section that only contains passes is not reporting a validation.

What follows from it for the results is bounded. The S&P 500 appears as a leg in five of the six notes, so if this surface carries error the error is widespread. But the direction is not determined: a vol fit slightly off in either direction moves a worst-of value in the opposite direction to the cheapest leg, and §7 shows that a far larger volatility perturbation than anything suggested here moves the values the wrong way to explain the gap.

5.5 The payoff, the matrix and the pipeline

Three further checks sit outside the surface machinery. Each note's payoff is tested against scenarios worked by hand, so that a coupon paid on a barrier touch, a call on the first eligible date and a principal loss through the trigger each produce the arithmetic they should rather than merely a plausible number. The correlation estimator is tested on a matrix constructed to be non-positive-semidefinite, confirming the repair of Higham (2002) fires and returns a valid matrix rather than failing quietly at the Cholesky step. And the six term sheets are tested against the preserved filings, so that a coupon rate or a trigger transcribed into the configuration cannot drift from the document it came from.

Reproducibility is checked by construction rather than asserted. A fresh copy of the repository holding only the shipped files recalibrates all five surfaces from the preserved option chains bit for bit, passes the test suite, and regenerates the report byte for byte.

6 Results

6.1 Model value against the disclosed value

Table 4 gives the result the rest of the paper is about. Every model value lands below the issuer's disclosed estimated value. Across the five notes whose barriers are covered by quoted strikes the shortfall runs from \$11.41 to \$27.92 per \$1,000 of principal, or 1.17 to 2.97 per cent of the disclosed figure; the note held out of that range is further below still, at \$48.72. Standard errors are between \$0.24 and \$0.37, so the differences are one to two orders of magnitude larger than the simulation noise around them.

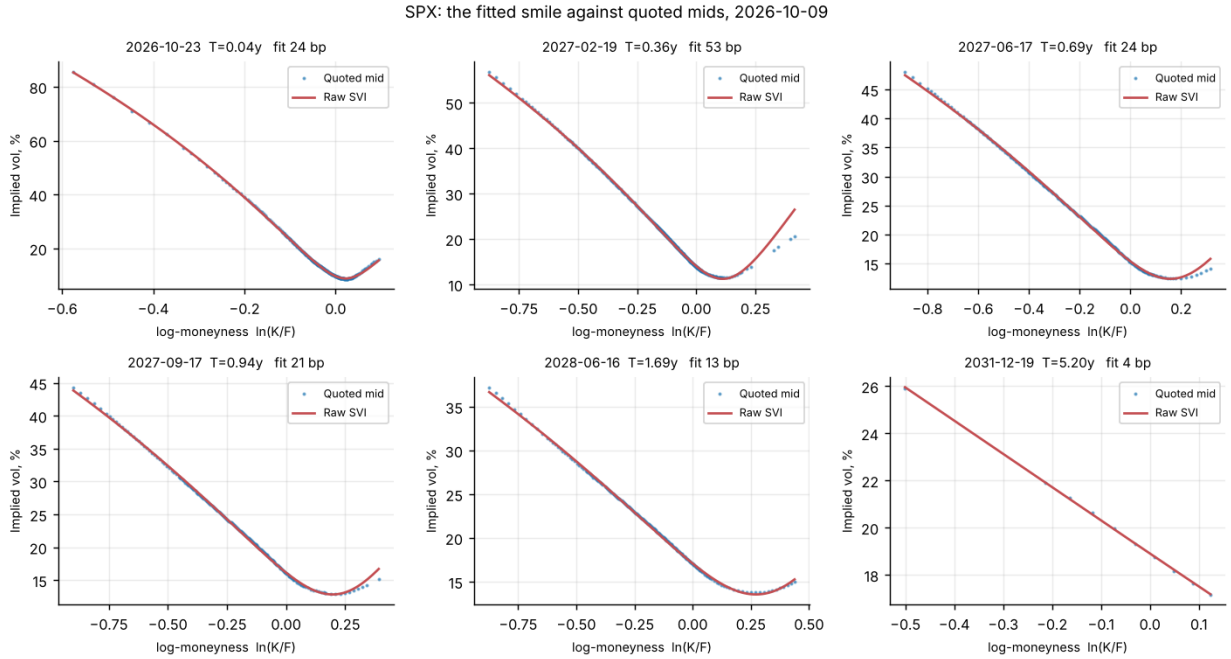


Figure 1. Fitted S&P 500 smiles against quoted mids, six expiries from the 9 October 2026 snapshot

Notes: Points are quoted mids, lines the fitted raw SVI slice. The figure in each panel title is that slice's root-mean-square fit error in basis points of implied volatility. Panels are ordered by maturity; the last is the longest listed S&P 500 expiry at 5.20 years, fitted from ten quoted volatility points, and each panel plots the quotes falling inside its own axis range rather than the whole fitted slice. Source: `scripts/make_figures.py`.

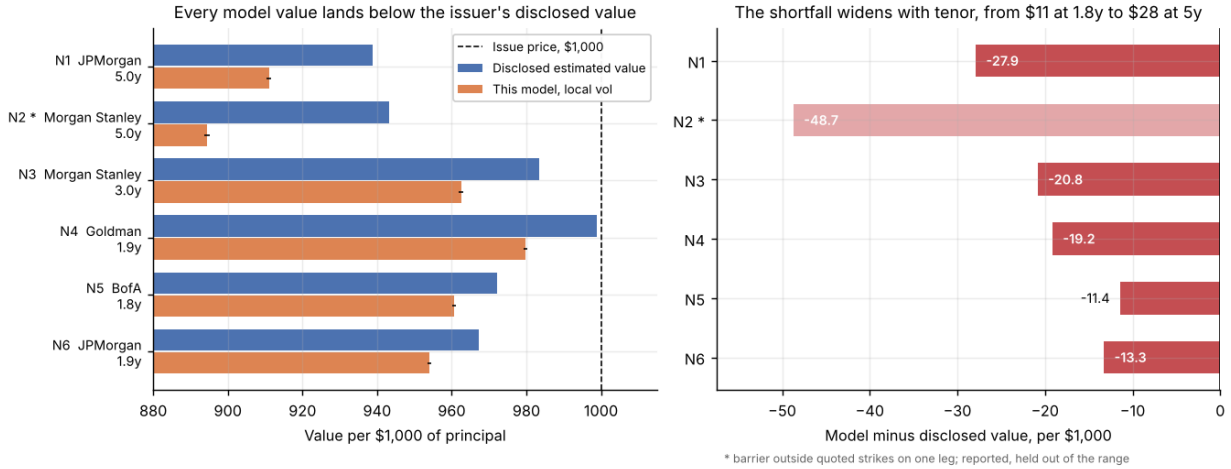


Figure 2. Model value against the disclosed estimated value, in levels and in differences, per $1,000 of principal

Notes: Left panel, the disclosed estimated value and the model value for each note, with the issue price marked; whiskers are one standard error of the simulated mean and are smaller than the bar width at this scale. Right panel, model value less disclosed value. The starred note is reported but held out of the ranges quoted in the text, for the barrier coverage reason given in §3. Source: `scripts/make_figures.py`.

Table 4. Model value against the disclosed estimated value, per $1,000 of principal

Note	Model value	s.e.	Disclosed	Gap	Gap (%)
N1	910.98	0.34	938.90	−27.92	−2.97
N2[a]	894.38	0.37	943.10	−48.72	−5.17
N3	962.56	0.29	983.40	−20.84	−2.12
N4	979.84	0.24	999.00	−19.16	−1.92
N5	960.69	0.26	972.10	−11.41	−1.17
N6	954.01	0.30	967.30	−13.29	−1.37

Notes: Model value is the mean discounted payoff over 200,000 paths with a European control variate, at a zero funding spread over the Treasury curve, with the issuer's call exercised by the Longstaff–Schwartz rule. The standard error is of that mean. Gap is model value less disclosed value, so a negative gap means the independent valuation lands below the figure the issuer discloses. Every note shows one. [a] Excluded from the ranges quoted in the text; see Table 2.

Table 5. The funding spread that would reconcile model and disclosed value, under two assumptions about the call

Note	Disclosed	Model	Spread (bp)	Model, no call	Spread (bp)
N1	938.90	910.98	-123	n/a	-1
N2	943.10	894.38	-204	n/a	-69
N3	983.40	962.56	-179	n/a	+76
N4	999.00	979.84	no solution	n/a	+99
N5	972.10	960.69	-278	n/a	n/a
N6	967.30	954.01	-156	n/a	+121

Notes: Each spread is the constant addition to the Treasury discount curve that sets the model value equal to the disclosed value, solved on a common set of paths. A funding spread should be positive and wider than the issuer's own senior bonds: the A-rated US corporate option-adjusted spread was 69 bp on the pricing date. Under optimal exercise every solved spread is negative, from -278 bp to -123 bp, which would require the issuer to fund below Treasuries. *No solution* means the disclosed value lies outside the bracket searched; §6 reports where that note's crossing lies. The auto-callable note has no no-call column: its redemption is contractual, so removing it describes a different security rather than a different modelling assumption.

Figure 2 shows the same thing twice, in levels and in differences. The ordering in the right panel is worth noting: among the five headline notes the shortfall increases with tenor, from $11.41 on the shortest at 1.75 years to $27.92 on the longest at 5.00. Five observations do not establish a relationship and nothing here tests one. But the shape is what a discount-rate difference would produce, since a constant spread compounds over a longer life, and that makes the sign of the next result more pointed than it would otherwise be.

6.2 The spread that would reconcile them

Solving (4) for the funding spread that sets the model value equal to the disclosed value gives Table 5. Under optimal exercise every solved spread is negative, from −278 to −123 basis points. The benchmark is 69 basis points, the option-adjusted spread on A-rated US corporate debt on the pricing date, and an issuer's own funding should sit above that rather than below it, since a structured note funds more dearly than a plain senior bond. The solved spreads are on the wrong side of zero, let alone of the benchmark.

One note returns no solution inside the bracket searched. Extending the grid for that note alone locates its crossing at −311 basis points, 11 basis points beyond the edge of the bracket, which confirms the absence of a solution was a property of the search range and not of the note. It changes nothing about the conclusion: a note whose model value meets its disclosed value only when the issuer is assumed to borrow 311 basis points below Treasuries is not reconciled by funding either.

Reading the sign is the whole point. Discounting a liability at a wider spread lowers its value, so if the disclosed figure were the model value computed with the issuer's higher internal funding rate, the solved spread would come out positive. It comes out negative on every note. Whatever accounts for the difference between these two valuations, a funding rate the filings decline to publish is not sufficient to be it, and that is the candidate a reader reaches for first because the filings themselves point there.

The columns for the no-call case show how much of this depends on the exercise assumption. Removing the issuer's call raises three of the model values above their disclosed figures, turning those spreads positive. The call, in other words, is the largest single lever in the model, which is why it gets the next subsection to itself rather than a sentence here.

6.3 How much of the call would have to be ignored

The call assumption can be posed as a quantity rather than a switch. Give the issuer the funding spread it actually pays, 69 basis points on the pricing date, and value each note twice: once with the call exercised optimally and once with no early call at all. The disclosed value's position between those two bounds is

$$\varphi = \frac{V_{\text{no call}}(s) - \text{EV}_{\text{disclosed}}}{V_{\text{no call}}(s) - V_{\text{optimal}}(s)}, \tag{5}$$

the fraction of the call's value that the disclosed figure appears to carry. A φ between zero and one says the disclosed value is consistent with the issuer pricing some of the call and not all of it. A negative φ says something stronger: the disclosed value sits above the note valued with no call whatsoever, so no treatment of the call, however generous to the investor, reaches it at that funding rate.

Three of the five notes to which this applies return φ inside the unit interval, at 0.02, 0.16 and 0.29. For those the disclosed value is consistent with an issuer recognising between two and twenty-nine per cent of what optimal exercise is worth, which is a coherent position: a desk may well value its own call conservatively, and the Morgan Stanley notes' disclosed rational-exercise rule is itself a weaker commitment than optimality.

The other two return -1.55 and -3.87. Their disclosed values exceed what the notes are worth with the issuer's call removed entirely and funding charged at 69 basis points. Both are the five-year notes. For these the call cannot be the explanation at any fraction, and the residual has to come from somewhere else in the model, from a different funding assumption than the corporate benchmark, or from the disclosed figure itself.

6.4 What the issuer's own numbers say

One part of this needs no model at all, only the two figures the issuer prints on the cover. The gap between the issue price and the disclosed estimated value is attributed in the filings to the underwriting discount, the expenses of creating and marketing the notes, and the internal transfer between the issuing entity and its distribution affiliate (§2). Subtracting the disclosed commission from that gap leaves what is available for the other two components.

On five of the six notes that residual is positive, between \$2.90 and \$20.65 per \$1,000. On the sixth it is $-$\$4.25. That note was sold at par with a disclosed estimated value of \$999 and an underwriting discount of 0.525 per cent, so net proceeds to the issuer were \$994.75 against a liability the issuer valued at \$999. The estimated value is quoted to the dollar on that cover rather than to the cent, so the figure carries about fifty cents of rounding either way, which does not change its sign.

Table 6. The note decomposed into the positions a desk would book, per $1,000 of principal

Note	Funding bond	Coupon strip	Worst-of put	Issuer's call	Without call	With call
N1	778.77	299.87	−135.72	−27.99	942.92	914.93
N2	778.77	291.16	−151.68	−22.34	918.25	895.91
N3	864.65	234.00	−94.29	−40.09	1,004.37	964.27
N4	913.12	171.01	−67.22	−36.38	1,016.92	980.54
N5	920.50	90.38	−102.08	51.99	908.81	960.80
N6	913.12	194.24	−119.14	−34.23	988.22	954.00

Notes: The investor is long a funding bond and a coupon strip and short a worst-of put, and signs are from the investor's side. Where redemption is the issuer's decision the call is a cost to the investor and the column is negative. On the one auto-callable note it is positive: redemption is mechanical, returns principal early in states where the note would otherwise stay exposed, and so favours the holder. *Without call* is the sum of the first three columns and *With call* subtracts the fourth. The decomposition runs on 50,000 paths with no control variate, because every part has to be simulated under the same draws for the parts to add up; Table 4 reports the headline estimator at 200,000 paths with the control variate. The last column is therefore a coarser estimate of the same quantity, not a competing one.

This is not necessarily an error. The internal transfer component can be negative if the issuing entity's funding benefit is recognised elsewhere, and nothing in the disclosure requires the three components to be individually positive. What it does show is that the disclosed numbers on that cover do not by themselves decompose into a positive margin, and that the arithmetic is available to any reader who adds two figures printed a few lines apart.

6.5 Where the value sits

Table 6 splits each note into the positions a desk would book: a funding bond for the principal, a strip of contingent coupons, a short worst-of put, and the issuer's call. The investor is long the first two and short the last two. The split is diagnostic rather than a result, but it locates the magnitudes. After the funding bond the coupon strip is the largest component on five of the six notes, and the short worst-of put the largest on the sixth, which is the note with the lowest coupon in the sample.

The call column needs reading with its sign. On the five notes where redemption is the issuer's decision it is a cost to the investor, between $22.34 and $40.09, and it is of the same order as the $11 to $28 to be explained, which is what makes (5) a real question rather than an arithmetic curiosity. On the auto-callable note it runs the other way, at $51.99 in the investor's favour: a mechanical redemption at 93.25 per cent of the initial level returns principal early in states where the note would otherwise remain exposed, and the holder gains from it. That is also why (5) is not computed for that note. Its redemption is a contractual term rather than a discretionary input, so there is no assumption about it to vary.

7 Which inputs can explain the difference

A candidate explanation has to clear two bars. It must move the model value toward the disclosed figure rather than away from it, and it must be capable of moving it far enough. Direction alone is not enough and magnitude alone is meaningless if the sign is wrong. Table 7 reports each input's measured effect on a common set of paths, with one input moved at a time and the rest held at their calibrated values. Taken in turn, four candidates fail one bar or the other and two survive both. A third survivor sits outside that table, because it is a dimension the model used here does not have.

Table 7. Candidate explanations for the gap, one input at a time, per $1,000 of principal

Note	Gap to close	+0.05 correlation	Flat vol	Timing	Call cost
N1	−27.92	9.14	62.49	−4.04	27.99
N2	−48.72	7.70	70.21	−7.94	22.34
N3	−20.84	5.58	44.50	−4.56	40.09
N4	−19.16	3.55	31.90	−3.64	36.38
N5	−11.41	3.32	6.65	−3.76	−51.99
N6	−13.29	3.10	26.71	−4.55	34.23

Notes: Each column is the change in model value from moving one input while the rest are held at their calibrated values, on a common set of paths. *+0.05 correlation* raises every pairwise correlation, confirming the investor is long correlation on every note. *Flat vol* replaces the fitted surface with a single at-the-money volatility per leg. It raises every value, and by more than the gap on five of the six, so the shape of the surface is the one input measured here whose effect both runs toward the disclosed value and is large enough to span the difference. *Timing* aligns the surface to the pricing date rather than the snapshot date and lowers every value, widening the gap. *Call cost* is what the issuer's option to redeem is worth to the issuer, from the decomposition in Table 6. No single column closes the gap in the first; §7 takes them in turn.

7.1 Ruled out by direction

Two candidates move the wrong way.

The funding rate was settled in §6 and is the first a reader reaches for, because the filings point there. Raising the discount spread lowers the model value and widens the gap; closing it requires a negative spread on every note. Nothing about a funding assumption the issuer declines to publish repairs that, because the sign is wrong before the magnitude is discussed.

Observation timing fails the same way. The surface is observed two business days after five of the notes priced, and aligning it back to each pricing date using the move in the matching volatility index lowers every value further, by $3.64 to $4.56 across the headline notes. The adjustment is real and it belongs in the model, but it enlarges the quantity to be explained rather than reducing it.

7.2 Right direction, too small

Two candidates push the right way and cannot reach.

The proxied Dow leg, discussed in §3, raises the affected note by $2.61 with a standard error of $0.73 when corrected on its own. That is about five per cent of that note's gap, and it touches one note of the six. It is a real bias with a measured size, and the size is not the size of the problem.

Extrapolating the surface past the longest listed expiry matters only where the extrapolation is long. Carrying the last observed variance slope forward instead of freezing the smile raises the five-year note by $12.36, which is 44 per cent of its gap and far from negligible. On the three notes under two years the same change is worth $0.22 to $0.77. The term structure assumption is therefore a genuine source of uncertainty on the long end of the sample and close to irrelevant on the short end, while the gap is present throughout. It cannot be the common explanation, though it plausibly contributes to the largest case.

7.3 Bounded by the exercise policy

Suboptimal exercise was quantified in §6 and the bound it gives is partial. On three notes the disclosed value sits between the no-call and optimally-called valuations, consistent with an issuer recognising two to

twenty-nine per cent of the call's worth. On the two five-year notes it sits above the no-call valuation entirely at a 69 basis point funding spread, so no exercise assumption reaches it. An explanation that works for three notes and is arithmetically unavailable for two others is not the explanation, although it may well be part of the answer for the three.

7.4 What survives

Two of the measured inputs clear both bars, and a third candidate clears them on published evidence without being measurable here at all. None of the three is published in the filings.

Correlation. The investor is long correlation on every note, so raising it raises the model value. The measured sensitivity is $3.10 to $9.14 per 0.05 of uniform pairwise correlation. Closing each gap on correlation alone would take a uniform increase of 0.15 to 0.27, which is three to five times the perturbation actually measured, obtained by extrapolating a local sensitivity well beyond the range over which it was computed and therefore an estimate of the order rather than a precise requirement. The direction is right and there is a reason to expect a difference: a desk selling a worst-of is short correlation and marks it above a realised estimate, which is the approximation §4 flagged as the method's largest. Whether the difference is of this size is not something six notes can settle. The candidate is live and unresolved.

The shape of the volatility surface. Replacing each leg's fitted surface with a single at-the-money volatility raises all six notes, by $6.65 to $70.21. This comparison is quoted over all six rather than over the headline five, because what it measures is a sensitivity rather than a level, and the barrier-coverage objection of §3 bears on the excluded note's value, not on how that value responds to flattening its surface. On five of the six the move is larger than the gap: a fully flat surface overshoots the disclosed value by $12.64 to $33.93, and falls short on only one note, by $4.87. Somewhere between the quoted smile and no smile at all there is a surface that reproduces each disclosed figure exactly.

The claim this supports is narrow and the narrowness matters. No desk prices a worst-of on a flat volatility surface, and nothing here suggests any issuer did. What the flat-vol valuation provides is a bound: it shows the distance between this paper's values and the disclosed ones lies inside the range that disagreement about the shape of the surface can generate, and comfortably inside it on five notes. A structure whose principal trigger sits 30 to 40 per cent below the initial level, with coupon barriers 25 to 35 per cent below it, is priced off the wing of the smile rather than its level, and the wing is where quotes thin out and modelling choices diverge most. Two careful desks working from the same quotes can differ there by more than the amounts in Table 4.

The dynamics of that surface. A third candidate does not appear in Table 7, because the model used here cannot produce it. Local volatility reproduces today's surface and fixes how it evolves; stochastic and stochastic-local volatility models add a vol-of-vol dimension that a European repricing gate cannot see. Farkas, Ferrari, and Ulrych (2022) put the point sharply for autocallable barrier reverse convertibles: the stochastic parameters of a Heston local-stochastic volatility model control the autocallable price *while leaving the fit to European options unaffected*, and the richer dynamics typically produce higher prices than local volatility. Higher is the direction that would close the gap measured here.

Two qualifications stop that from being an answer. Their products are single-asset and these are worst-of on three underlyings, so the direction is suggestive rather than transferable. And the candidates are not independent: De Col and Kuppinger (2017) study equity worst-of autocallables under multi-dimensional

stochastic local volatility and find the correlations between the stochastic variances central to pricing them, so correlation and volatility dynamics interact in precisely this product class rather than contributing separately. Setting them out as two columns of a table, as this section necessarily does, is itself a simplification.

The consequence for §5 is worth stating plainly. That section establishes the engine returns the vanilla prices it was calibrated to. Farkas et al. (2022) establish that two models can agree on every one of those prices and still disagree on an autocallable. The validation gate is silent on this dimension by construction, not by oversight.

7.5 Taken together

Of the inputs this study can measure, none closes the gap on every note while moving in the right direction except the shape of the volatility surface, and that one does so by bounding the problem rather than by identifying a choice. Funding and timing are excluded by sign. The proxy and the extrapolation are too small or too narrow. Exercise policy accounts for part of three notes and none of two. Correlation remains, sufficient in principle at a shift this sample cannot adjudicate. Beyond what is measured, the dynamics of the surface remain open, and the published direction favours them.

That is a weaker conclusion than naming a cause and a stronger one than reporting an unexplained residual. All three surviving candidates are inputs the disclosure does not contain, and one of them, correlation, is absent even from the lists of principal inputs that two of these issuers print (§2). A reader holding the pricing supplement cannot distinguish among them, which is a statement about what the disclosure makes checkable rather than about what any issuer did.

8 Limitations

8.1 Inputs that are not what a desk would use

The volatility surface is observed two trading days after five of the six notes priced, which is a consequence of the filing window in §3 rather than a choice. §7 measures the cost of that gap, note by note, and it runs the wrong way to help: correcting for it lowers values further.

Correlation is realised rather than implied, for the reason given in §4: vanilla options carry no co-movement information and no free source of implied correlation exists. The bias runs in a known direction, and §7 treats it as one of the surviving candidates rather than burying it here.

Discounting uses the Treasury par yield curve where a dealer would use SOFR OIS, again because no free source of OIS quotes exists. The two differ by a handful of basis points at these tenors. That is small against funding spreads measured in the hundreds, and it is not nothing.

8.2 The Dow leg, and what the coverage test does not promise

One leg is priced off fund options standing in for an index (§3). Two problems follow, and the second is worse than the first. The level bias is measured at 0.52 volatility points and is worth $2.61 on the affected note. The more serious problem is that the fund's long-dated quotes never reach the strike where that note's barrier sits, so for the whole of its life past the first year the downside is priced off an extrapolation rather than a quote. That note is reported throughout and excluded from every quoted range, and its $0.37 standard error understates its true uncertainty by an amount this study cannot quantify.

The coverage test that produced that exclusion is a threshold, not a guarantee. It asks whether a quoted strike exists at or below the trigger on fitted slices past one year, and a thin market can satisfy it on a single strike while still pricing the barrier off very little. The other five notes reference index options deep enough that this is unlikely. Nothing here rules it out. Appendix C gives the counts leg by leg so a reader can apply a stricter threshold.

8.3 What local volatility does not do

The engine is calibrated to European option prices and reproduces them, within the bounds §5 reports. These notes are path-dependent, and nothing in that validation establishes that local volatility prices the path dependence correctly. It establishes that the terminal distributions are right.

This is the model's known weakness rather than an incidental one. Local volatility reproduces today's surface exactly and misgoverns how that surface evolves, flattening forward skew in a way that stochastic and stochastic-local volatility models are built to repair (Qu, 2016). It remains market practice for exotics of this shape because it handles multi-dimensionality, where stochastic volatility models become expensive to calibrate across several correlated underlyings (Grzelak, 2019). For a worst-of autocallable, whose value depends on the joint distribution at every observation date and not only at maturity, the dynamics are a real exposure.

The flat-volatility comparison in §7 bounds how much the *shape* of the surface matters. It does not bound how much the dynamics of that shape matter, and no result in this paper does. What can be said is that the direction is not unknown: Farkas et al. (2022) report that a Heston local-stochastic volatility model prices autocallables above local volatility, with the stochastic parameters moving the autocallable price while the fit to European options is unchanged. Their products are single-asset. Whether the same holds for a worst-of on three underlyings, where De Col and Kuppinger (2017) find the correlations between stochastic variances to be central, is not something this paper establishes.

8.4 Exercise, precision and sample

The issuer's call is solved under optimal exercise, and Longstaff–Schwartz with a finite basis returns a policy that is suboptimal to some degree. That error biases the investor's value upward relative to true optimality, so it pushes the model value toward the disclosed figure and cannot account for a model that is already too low. The direction is favourable to the finding, which is worth stating explicitly rather than leaving as an unexamined approximation.

One issuer discloses its estimated value to the nearest dollar rather than the cent, so the comparison for that note carries a dollar of irreducible disclosure precision against a gap of $19.16. It does not affect the sign of anything reported.

Six notes, four issuers, one pricing week. Every number here is a case study and none is an estimate of a population parameter. The 129 final supplements identified for that week would support a cross-sectional analysis of the disclosed margin against tenor, coupon, barrier distance and the number of underlyings. That is left undone, and it is the obvious next thing to do.

9 Conclusion

No. An independent valuation built from public option quotes, a public rates curve and the terms in the filing itself does not reproduce the issuer's disclosed estimated value. It lands below it on all six notes, by $11.41 to $27.92 per $1,000 across the five whose barriers are covered by quoted strikes, against simulation standard errors between $0.24 and $0.37.

The funding rate, which the filings themselves name as the input the reader cannot see, does not account for it. Solving for the spread that would reconcile the two valuations returns a negative number on every note, from −278 to −123 basis points against an A-rated corporate benchmark of 69. The sign is what carries the result: a disclosed value computed with a higher internal funding rate would reconcile at a positive spread, and none of them do.

Taking the remaining candidates one at a time excludes four more. Observation timing moves values further away. The proxied Dow leg and the choice of term extrapolation move the right way but are too small, or matter on too few notes, to be the common cause. Suboptimal exercise accounts for part of three notes and is arithmetically unavailable for the two five-year ones, whose disclosed values sit above what the notes are worth with the issuer's call removed altogether.

Two measured candidates survive on both direction and magnitude. The correlation assumption would close each gap with a uniform increase of 0.15 to 0.27, which is large but runs the way a dealer's mark is known to run against a realised estimate. The shape of the volatility surface would close it comfortably: a fully flat surface overshoots the disclosed value on five of the six notes. A third candidate cannot be measured with a local-volatility model at all, and the published direction favours it: Farkas et al. (2022) report that richer volatility dynamics price autocallables above local volatility, moving the price while leaving the fit to European options untouched. None of the three is published in the filing, and one of them, correlation, is not even named in the lists of principal valuation inputs that two of these issuers print.

What this paper contributes, and how much

Two things, both narrower than the subject they sit in.

The first is a verification exercise on the product class the nearest published work set aside. Bauer et al. (2020) reproduce issuer-disclosed values for German retail certificates and leave autocallables to later work; this paper prices six of them from public inputs and reports the funding-spread inversion as a diagnostic, which converts an unexplained difference into a statement about what the difference cannot be. The result is a case study on six notes from one week, not an estimate of anything about the market. Whether these gaps are typical is unknown, and nothing here should be read as suggesting they are.

The second is a census of what the exercise can reach at all. Of 1,086 supplements filed on Form 424B2 in the sample week, 129 are final worst-of notes with a contingent coupon and a disclosed estimated value, and 71 of those can be priced from public option quotes. A reader checking this disclosure can therefore reach 55 per cent of the qualifying market, and the inaccessible remainder is not a random sample of it: 33 of the 42 notes excluded for unlisted options reference a single US sector index that has no options market of its own. That bound is a fact about the disclosure regime rather than about any issuer, and it would hold for any outside party attempting the same check.

What would settle it

None of the three surviving candidates requires access to an issuer's models to test. Two of them, though, are not cleanly separable from each other, which shapes the order the tests should be run in.

Correlation could be tested directly with dispersion trade data or with quotes on multi-asset structures from which an implied correlation can be backed out. If dealer correlation marks on these underlyings sit 0.15 to 0.27 above the realised estimate over the relevant horizon, correlation is sufficient on its own and the question becomes how much of the gap it takes. If they sit much closer, it is not.

The surface dynamics could be tested by repricing the same six notes under a stochastic-local volatility model calibrated to the same quotes. The single-asset version of that test has been run: Farkas et al. (2022) find a Heston local-stochastic volatility model prices autocallables above local volatility, with the stochastic parameters moving the price while the European fit is untouched. What is missing is the multi-asset case, where De Col and Kuppinger (2017) find the correlations between stochastic variances to be central, and where the dynamics candidate and the correlation candidate stop being separable. Running it on these six notes would show whether the residual reported here survives a model that can see the dimension this one cannot.

The sample is the third thing to extend, and the cheapest. The 129 final supplements identified for one week would support a cross-sectional analysis of the disclosed margin against tenor, coupon, barrier distance and the number of underlyings, and the screening code that produced them runs forward on any later week. Whether the pattern reported here survives contact with a hundred notes rather than six is the question this paper cannot answer and the one most worth asking next.

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Disclosure statement

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Data availability statement

Every input to this study is public. The six pricing supplements are preserved in full under `references/sec_filings/` and remain retrievable from EDGAR under the accession numbers recorded in `references/source_register.csv`. Option quotes are delayed end-of-day chains from Cboe's public endpoint, captured on 9 October 2026. The eight chains the reported results depend on are tracked in the repository alongside the SHA-256 digest of each file, so a reader can confirm that the surfaces were fitted to the quotes recorded here rather than to a later capture. The other chains taken that morning are not tracked, and since the endpoint serves only current quotes they cannot be retrieved for that date afterwards. The Treasury par yield curve, the Cboe volatility index histories and the FRED series are published by the issuing institution and are downloaded by `scripts/fetch_reference_data.py`. Daily closes for the one exchange-traded fund among the underlyings come from a commercial endpoint rather than an official series, which is noted where that note is priced.

Code availability

The pricing engine, the calibration code, the test suite and the frozen outputs behind every number reported here are available at `https://github.com/soham-parundekar/autocallables-notes-pricing`. The Monte Carlo results reproduce exactly rather than approximately: the path count and the seed are recorded in `config/params.yaml`, and `make surfaces validate analysis report` regenerates the calibrated surfaces, the tables and the figures from the preserved snapshot. The repository was cloned into a clean environment and that sequence rerun before the results were frozen, which returned the calibrated parameters to the last digit and the reported tables byte for byte.

Declaration of generative AI use

During the preparation of this work, the author used Claude in order to revise and restructure the manuscript, prepare the plotting script, literature checks and LaTeX preparation. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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A Term sheet detail for the six notes

Each note is identified below by CUSIP and by the EDGAR accession number of the priced supplement it was read from. All six documents are preserved in full in the project repository, and all six remain retrievable

from EDGAR under these accession numbers. The economic terms appear in Table 2; what follows is the information needed to locate the source and to see which structural variant each note is.

Table 8. Filing identifiers and structural variant

Note	CUSIP	Accession number	Coupon frequency	Redemption
N1	46661PD38	0001918704-26-031125	monthly	issuer's option
N2	61781LWY2	0001839882-26-050390	monthly	rational-exercise rule
N3	61781LM23	0001839882-26-050393	monthly	rational-exercise rule
N4	40058M7F9	0001193125-26-419171	monthly	issuer's option
N5	09712CWN9	0001918704-26-030916	quarterly	automatic at 93.25%
N6	46661PL96	0001918704-26-031101	monthly	issuer's option

Notes: Product names as printed on the covers are *Callable Contingent Interest Notes* (N1, N6), *Callable Contingent Income Securities* (N2, N3), *Callable Contingent Coupon Index-Linked Notes* (N4) and *Contingent Income Auto-Callable Yield Notes* (N5). Each note is issued by a finance subsidiary and guaranteed by its parent. The three notes redeemable at the issuer's option become redeemable on scheduled payment dates after an initial non-call period: from the twelfth payment date on N1 and from the third on N4 and N6, in each case excluding the final payment. The two governed by the rational-exercise rule are redeemable on every stated redemption date, 48 of them on N2 and 30 on N3.

One structural difference matters for the simulation and is not visible in Table 2. Five of the notes observe the downside trigger only at maturity, so a breach is a European condition on the terminal level. The auto-callable note observes it on any trading day, which is an American knock-in and is handled in the simulation by a per-path indicator updated at every daily step rather than at the observation dates alone.

B SVI calibration and the arbitrage conditions

The fit

Fitting all five raw parameters of (1) at once is possible and unreliable. The objective has local minima, and on the S&P 500 expiry at $T = 0.36$ a direct least-squares run in this sample converged to $\rho = +0.39$, a positive skew, which is the wrong sign for an equity index and left 26 basis points of error at the money.

The fit used here is the quasi-explicit reduction. The reduction itself predates the arbitrage-free treatment, having been proposed in a 2009 whitepaper that Martini and Mingone (2022) record as its origin; it is their paper that characterises which reduced parameters correspond to an arbitrage-free slice, which is what makes a constrained refit well posed. Substituting $y = (k - m)/s$ turns (1) into

$$w(y) = a + (b\,\rho\,s)\,y + (b\,s)\sqrt{y^2 + 1}, \tag{6}$$

which is linear in the three coefficients $(a,\ c = b\rho s,\ d = bs)$. The outer search therefore has to handle only m and s, with the inner problem solved exactly as a weighted linear least squares at each candidate pair. Two constraints make the solution a valid slice: $d \geq |c|$, which is $|\rho| \leq 1$, and $a + \sqrt{d^2 - c^2} \geq 0$, which keeps total variance non-negative everywhere. The unconstrained inner solution usually satisfies both, and the constrained solver runs only when it does not, which is what keeps the two-dimensional outer search affordable.

The weighting, and why it is not uniform

Residuals are measured in total variance, but the quantity that matters is error in implied volatility, and the two objectives are not the same. On the fitted S&P 500 slice at $T = 0.94$, total variance runs from 0.0245 at the money to 0.1546 at the deepest quoted strike, a factor of 6.3. Least squares weights by the square of the residual, so an unweighted fit in w is dominated by the wing at roughly 40 to 1 and will fit the tail at the expense of the strikes that matter most for a note struck near the money.

Since $dw = 2\sigma T\, d\sigma$, scaling each weight by $1/(2\sigma T)^2$ converts the objective into squared volatility error while leaving it linear in the coefficients, which is what the reduction in (6) requires. That factor multiplies a base weight taken from quote tightness, the reciprocal of the relative bid-ask, capped at 50 so that one unusually tight quote cannot dominate its slice.

Enforcing the two conditions

The butterfly condition (2) is evaluated after each fit on a grid scoped to that expiry's own quoted strikes, padded by 0.10 in log-moneyness and capped at five standard deviations of total volatility. A slice violating it anywhere on that grid is refitted with the condition imposed, rather than having the offending wing smoothed afterwards.

The calendar condition is enforced pairwise between adjacent fitted expiries on the grid they share. Where total variance decreases in maturity at any point, the later slice is refitted subject to $w_{\text{later}}(k) \geq w_{\text{earlier}}(k)$. Expiries closer than 25 days apart are thinned beforehand, because two nearly coincident expiries divide independent fit error by a very small ΔT and manufacture violations that are an artefact of the spacing rather than a property of the surface.

Grid scope is not a detail. Enforcing these conditions on a single wide grid spanning $k \in [-1.5, 1.0]$ asks a two-week slice about strikes far beyond anything it quotes, where it is purely extrapolating, and the calendar condition then makes that extrapolated wing a floor for every later expiry. The constraint propagates forward from the slice with the least information rather than the most, and the degradation compounds across the term structure. Scoped instead to the strikes each pair of adjacent expiries actually quotes, the surfaces fit without that pressure, and the audit in §5 returns zero failures of either condition across all 68 slices.

C Quoted strike coverage at each note's barrier

A worst-of note's entire downside sits at its trigger, so the surface is queried there far more consequentially than at the money. Whether a quote exists at that point separates interpolating the smile from extrapolating it, and the distinction decides which notes carry ranges quoted in the text.

Table 9 counts, for every leg of every note, the fitted slices with maturity beyond one year whose quoted strikes reach down to the note's trigger. One year is the threshold because the short end of every surface here is quoted deep enough for the question not to arise. A leg scoring zero prices that note's downside, for the whole of its life past the first year, from an extrapolated wing.

One leg of one note scores zero. That note is reported throughout and held out of every range quoted in §6 and §7. The rule is stated as a count rather than a verdict so that a reader who considers the threshold too lenient can apply a stricter one without re-running anything: requiring coverage on a majority of deep slices rather than on one, for instance, changes no note's status in this sample.

Table 9. Quoted strike coverage at each note's downside trigger, by leg

Note	k at trigger	Covered slices past one year, by leg	Fully covered
N1	−0.431	NDX 2/2, RUT 3/3, SPX 8/8	yes
N2	−0.431	DJX 0/5, NDX 2/2, RUT 3/3	**no**
N3	−0.511	NDX 2/2, RUT 3/3, SPX 8/8	yes
N4	−0.511	NDX 2/2, RUT 3/3, SPX 8/8	yes
N5	−0.357	NDX 2/2, RUT 3/3, SPX 8/8	yes
N6	−0.511	RUT 3/3, SMH 5/5, SPX 8/8	yes

Notes: The trigger is expressed as log-moneyness relative to the initial level, so −0.431 is a trigger at 65 per cent of it. For each leg the entry counts the fitted slices with maturity beyond one year whose quoted strikes reach down to that level, against the number of such slices. A leg at 0 prices that note's entire downside past the first year from an extrapolated wing rather than from a quote. One note fails on one leg and is held out of every range quoted in the text, though its own figures are reported throughout.